ohm Technische Hochschule Nürnberg

IMU Gesture Glove

Jan Tischner

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02/07/2026

Data Fusion, Prof. Dr. Cristian Axenie

GESTURE GLOVE REPO

https://github.com/printjan/GestureGlove

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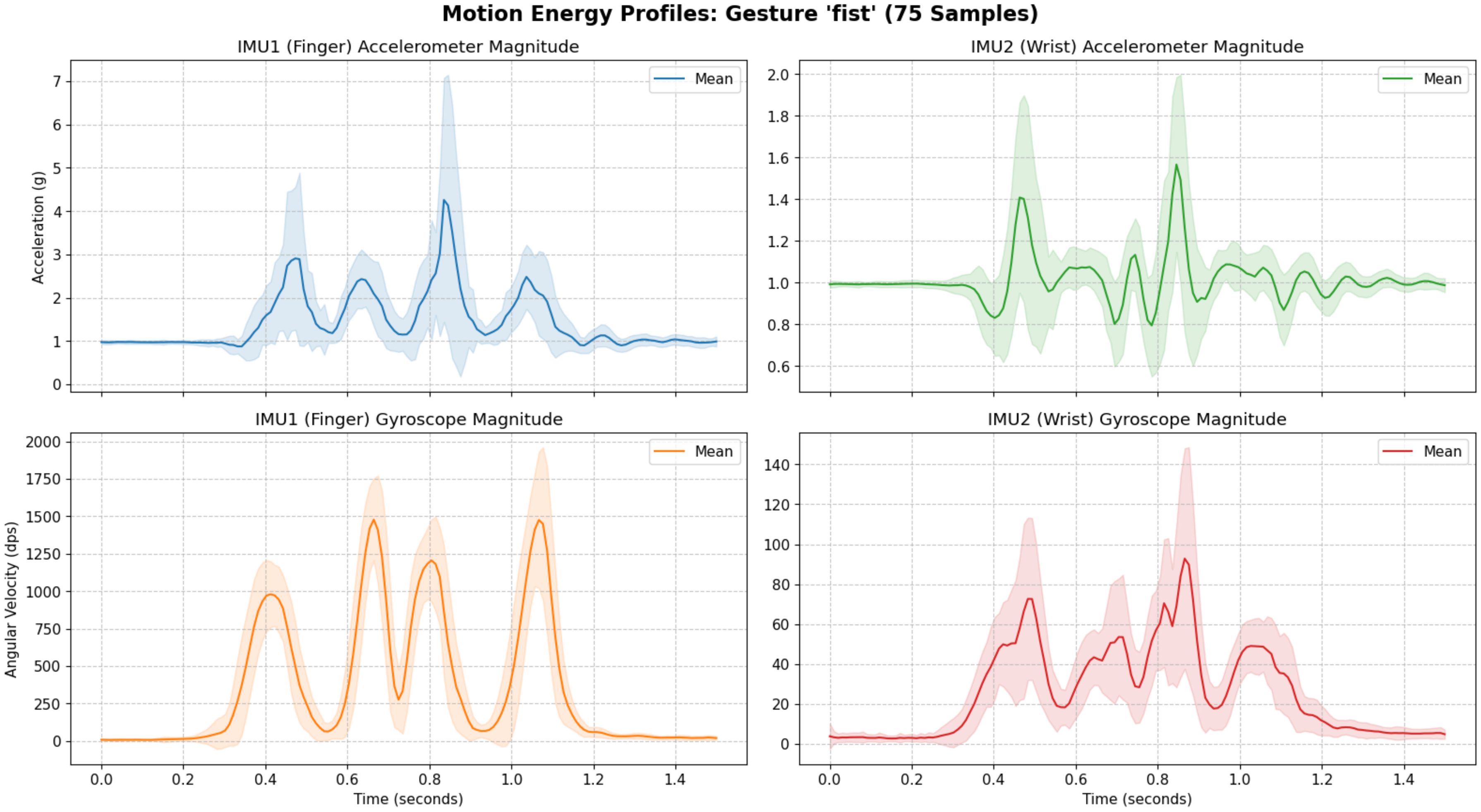
Gesture Recognition With IMU Data

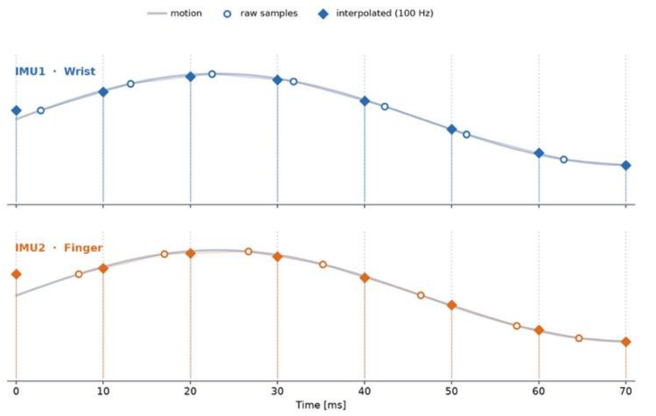
Project Idea: Control PowerPoint by recognizing gestures using an IMU glove.

Dual-IMU hardware setup:
- wrist captures arm translation,
- finger captures fine-grained rotation

Gesture Recognition With IMU Data

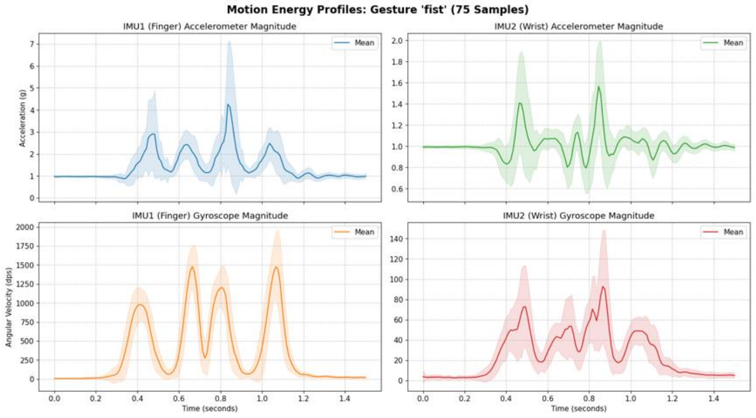
Implementation: Data Recording.

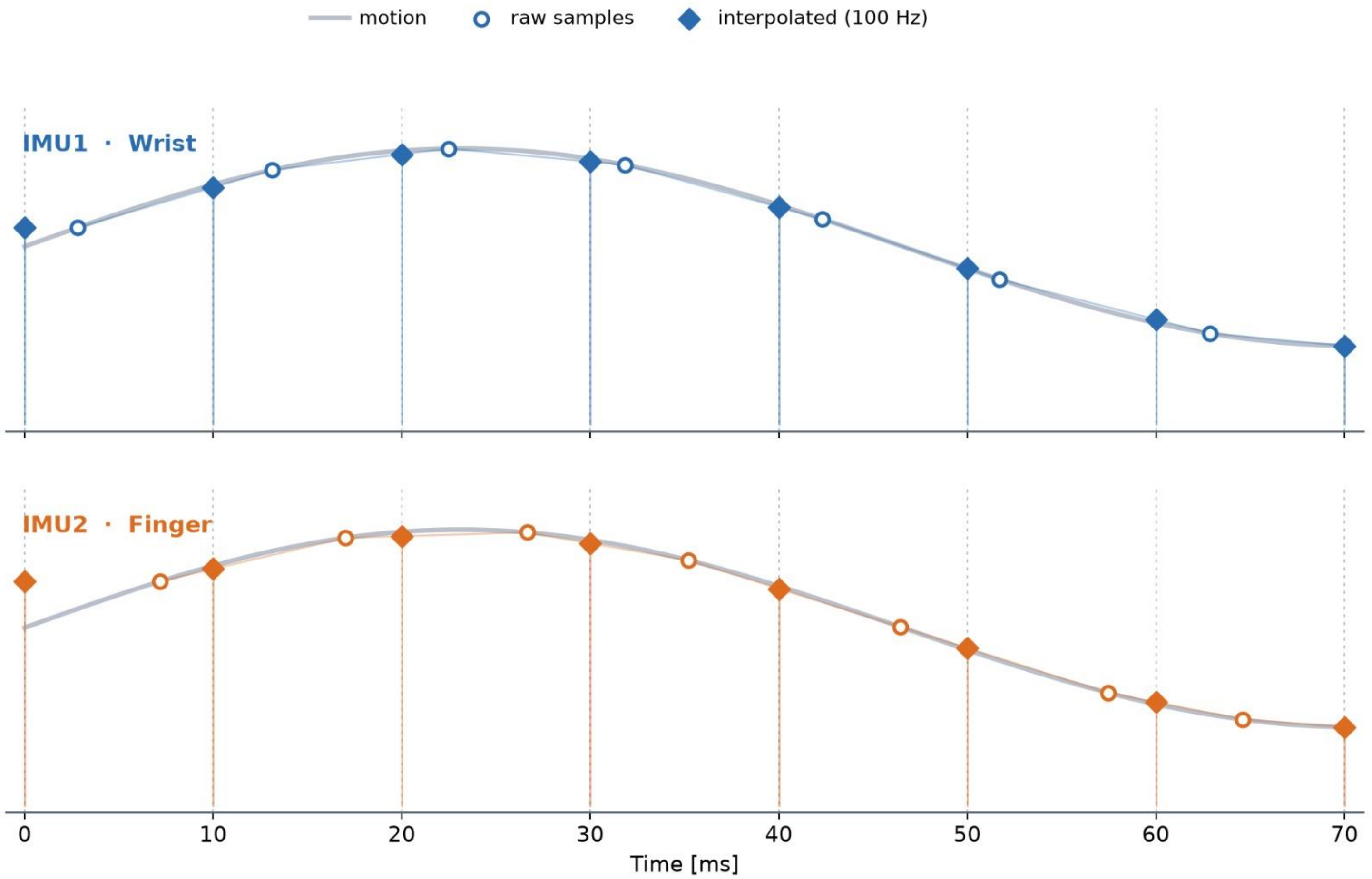




Gesture Recognition With IMU Data

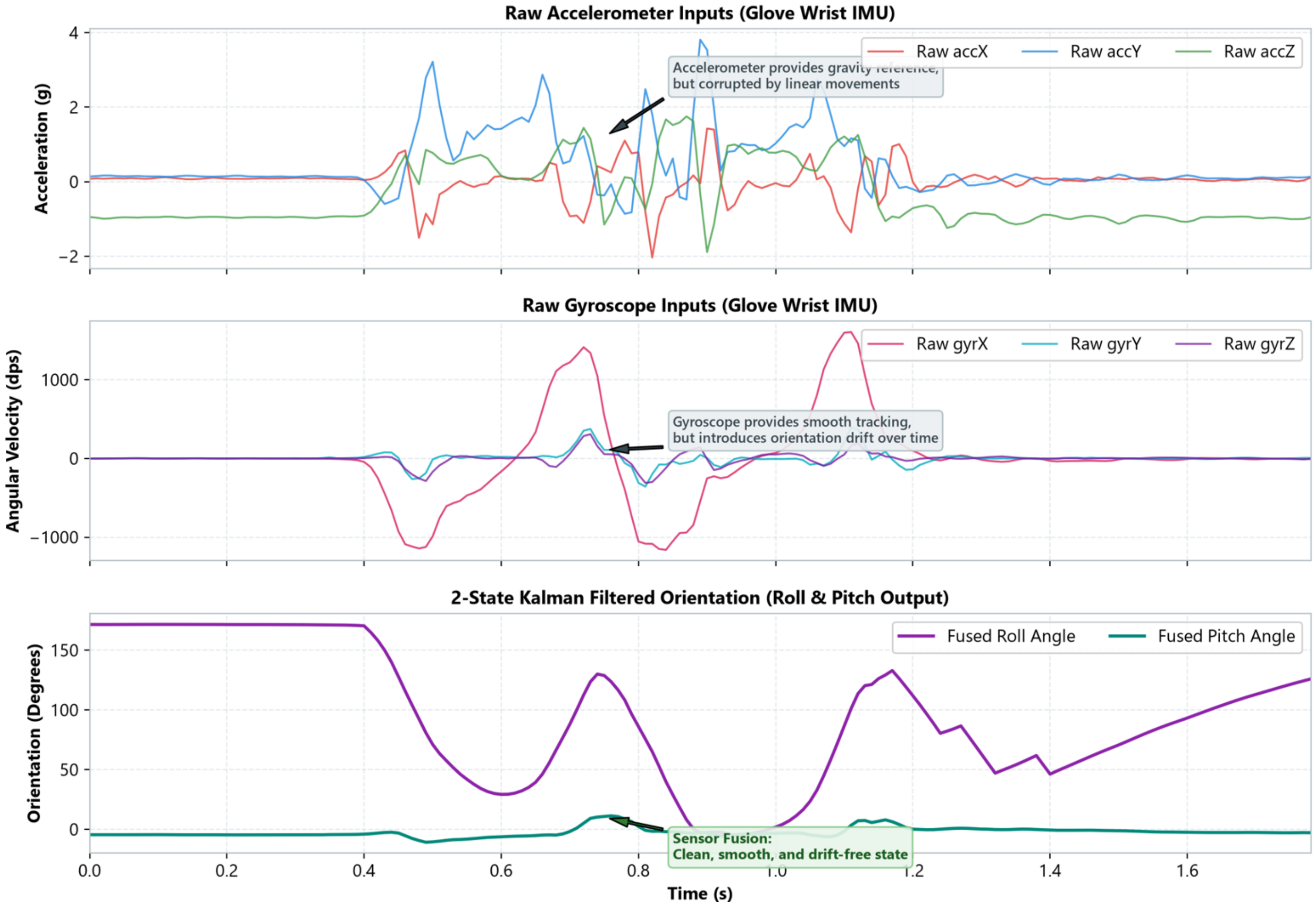
Implementation: Data Recording.

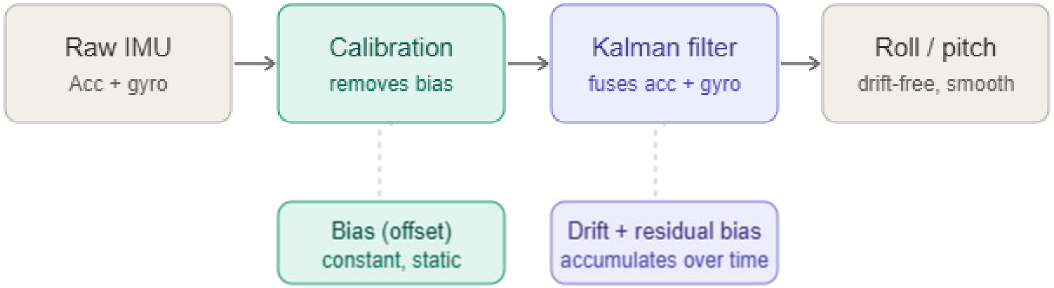




Gesture Recognition With IMU Data

Implementation: Filtering.





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Gesture Recognition With IMU Data

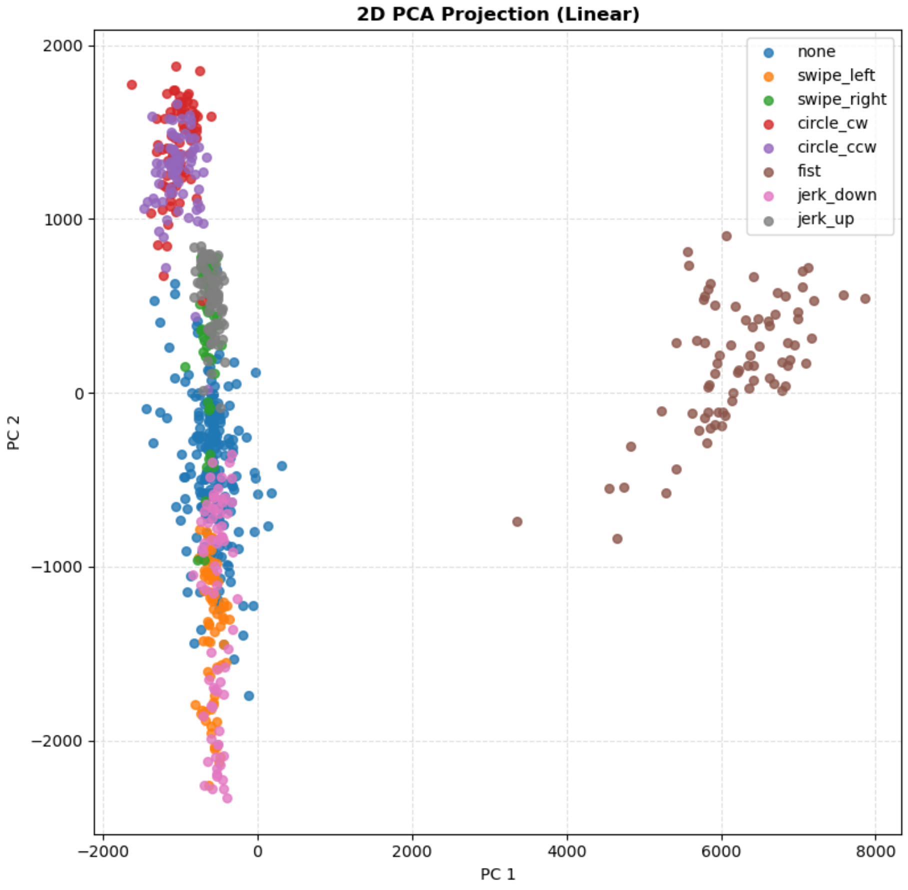
Implementation: Data Audit - Is this even implementable?

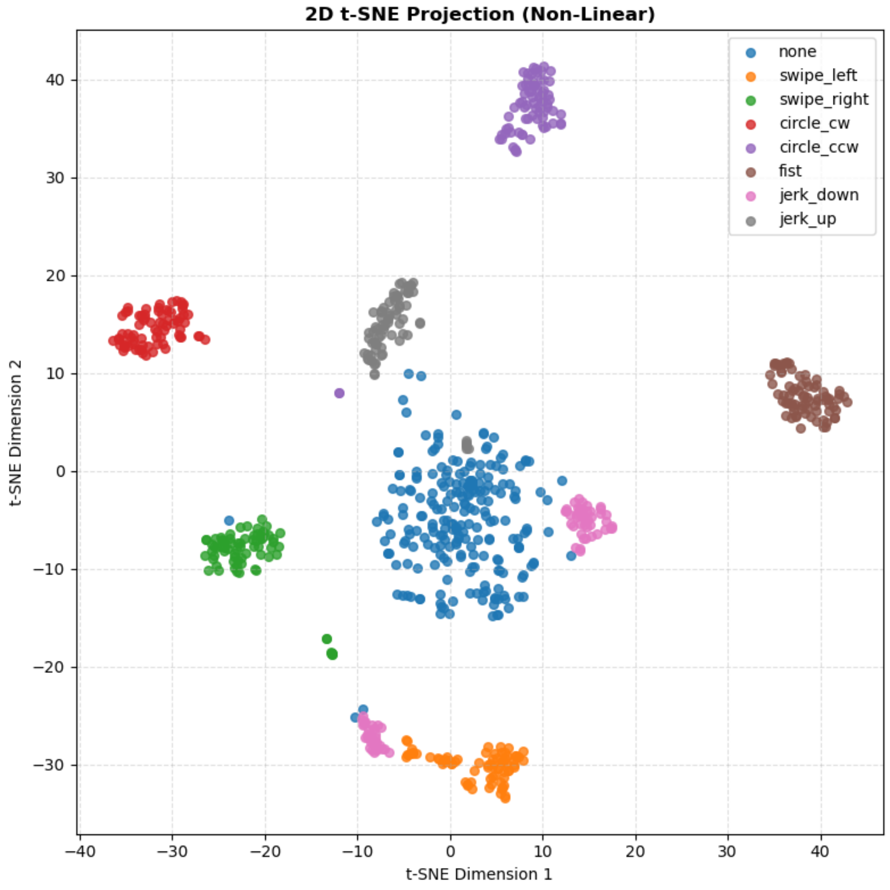
- **Linear Projection (PCA):**

  no clusters at all.

- **Non-Linear Manifold (t-SNE):**

  visible clustering.





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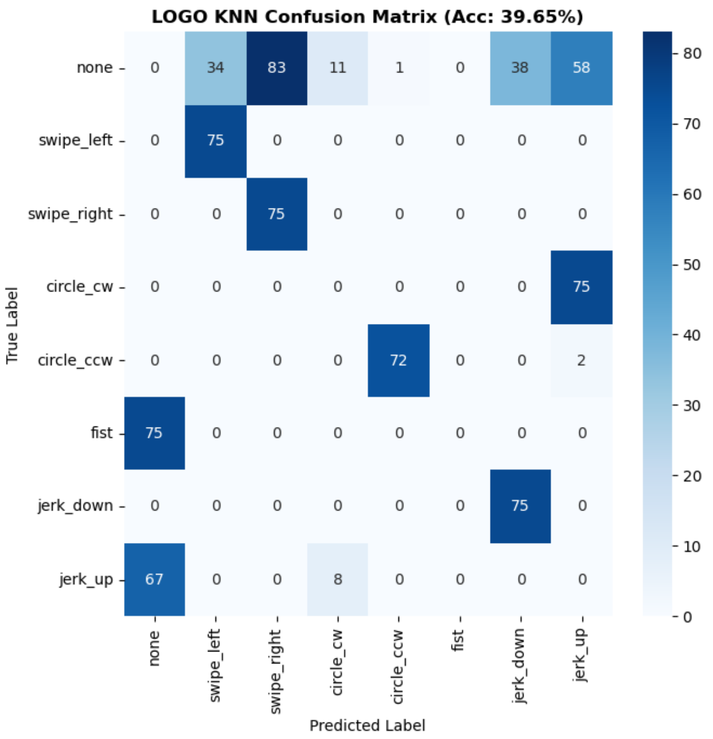
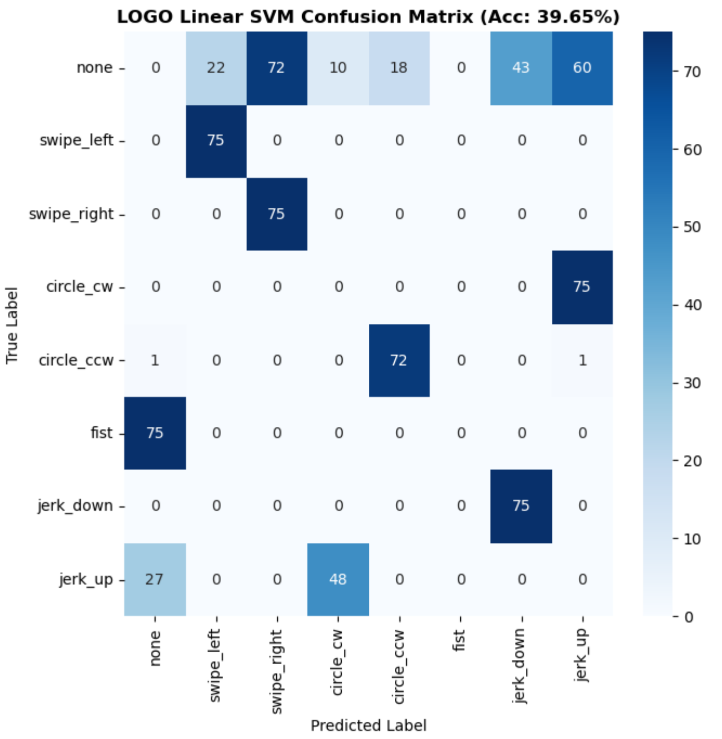
Gesture Recognition With IMU Data

Implementation: Data Audit - Is this even implementable?

- **KNN/SVM baseline:**

  only 39.6% accuracy





Conclusion: The data is separable, but only with models that learn temporal and spatial invariances — we need **deep learning**.

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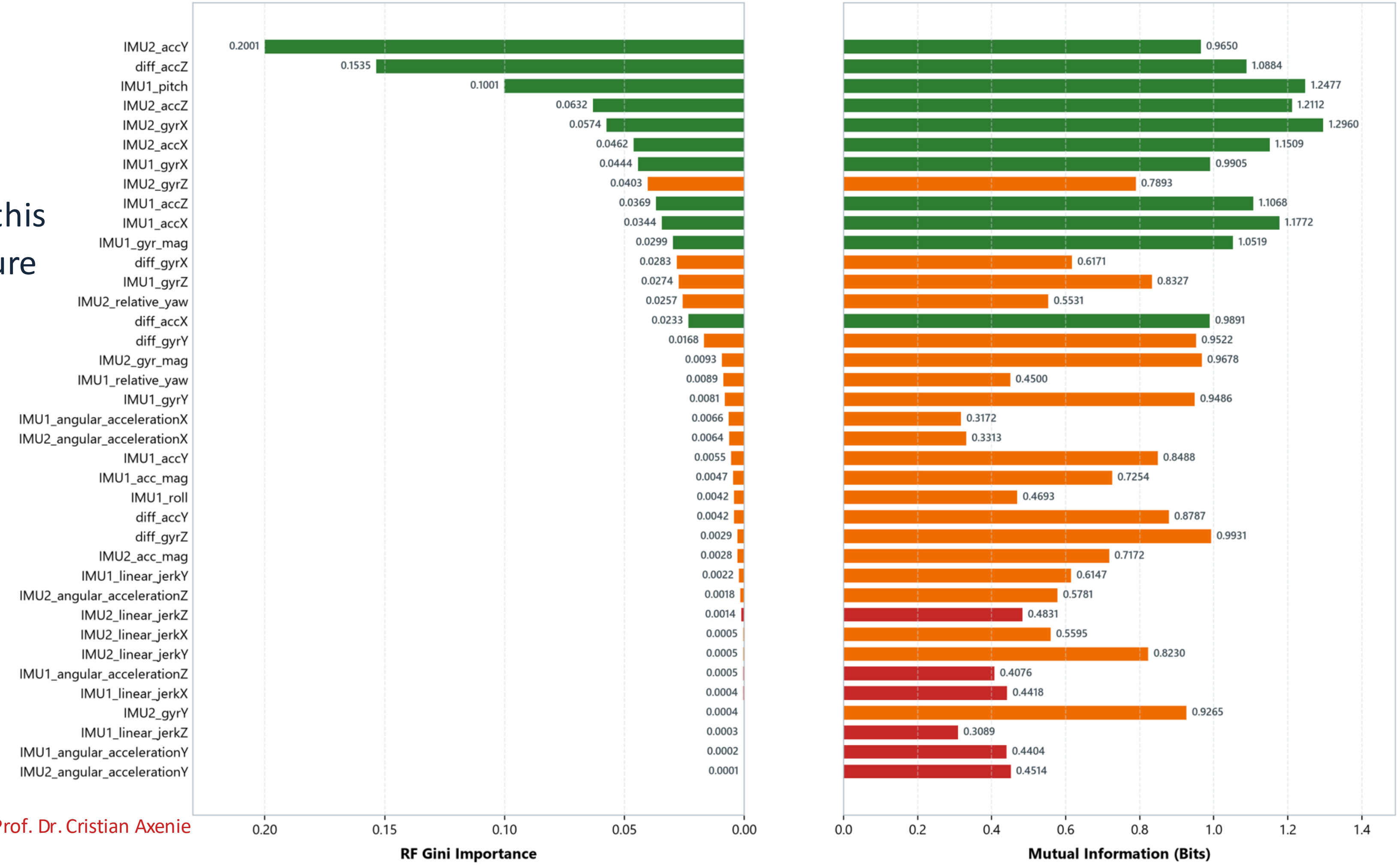
Gesture Recognition With IMU Data

Implementation: Data Audit – Feature Engineering.

Two metrics per feature:

- **Mutual Information (MI):**
  how much information does this channel carry about the gesture label?
- **Random Forest Gini Importance (RF)**
  how much does this channel contribute to decision boundary splits?



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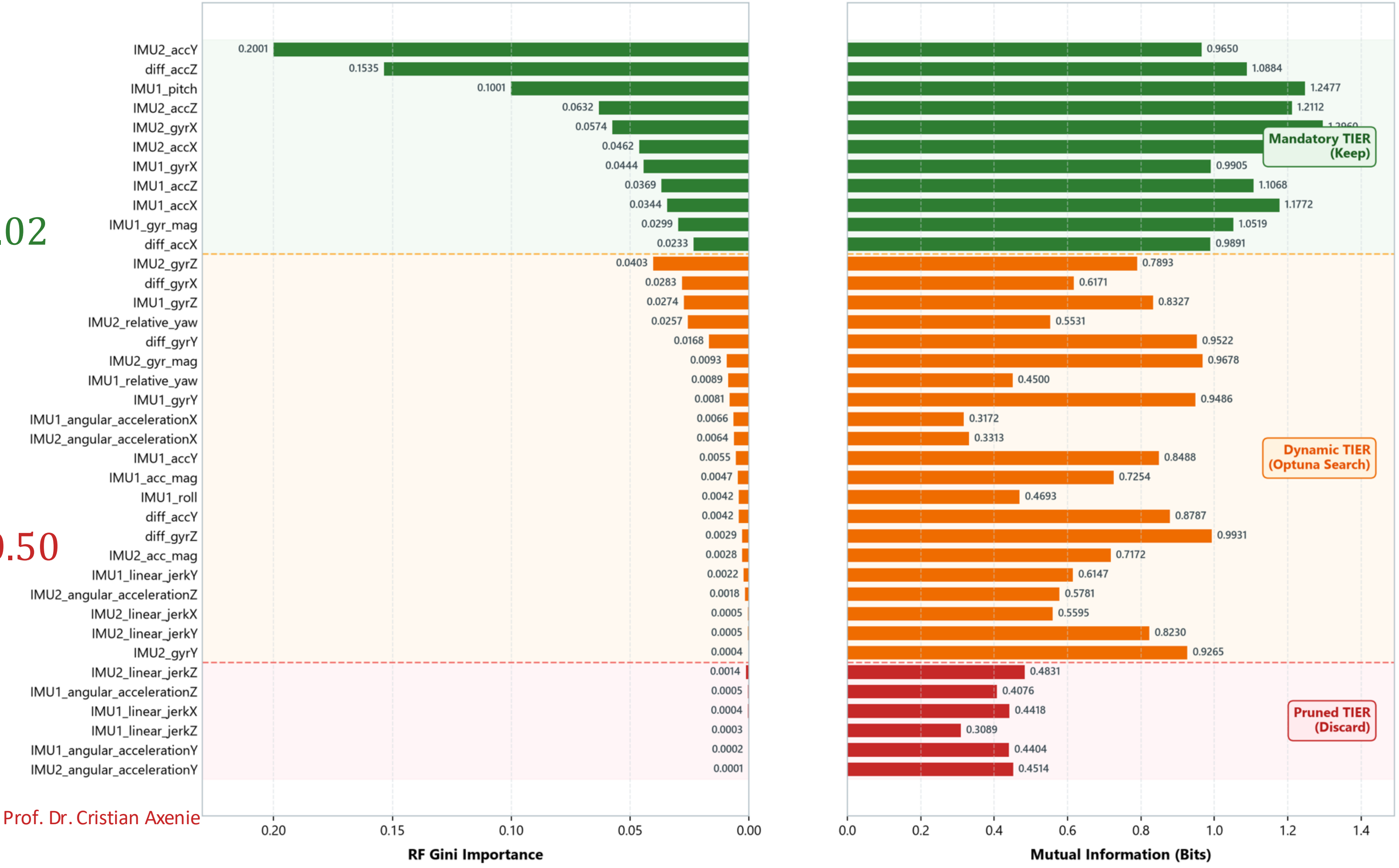
Gesture Recognition With IMU Data

Implementation: Data Audit – Feature Engineering.

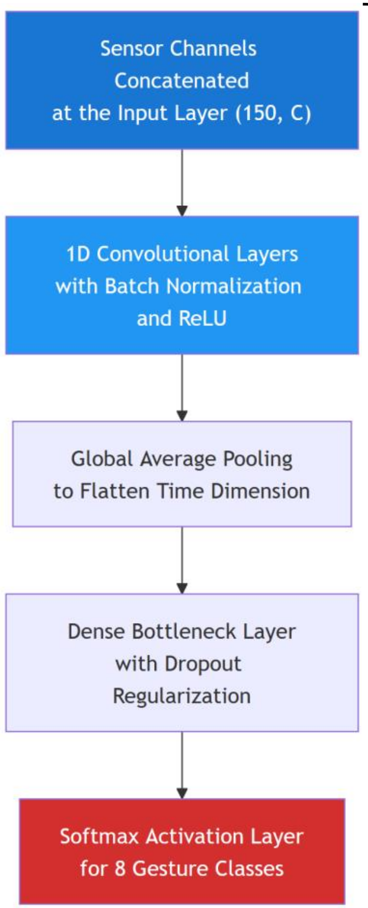
3-tier categorization:

- **Mandatory:**
  $MI > 0.90 \ \& \ RF > 0.02$
- Dynamic selection.
- **Pruned:**
  $RF < 0.002 \ \& \ MI < 0.50$

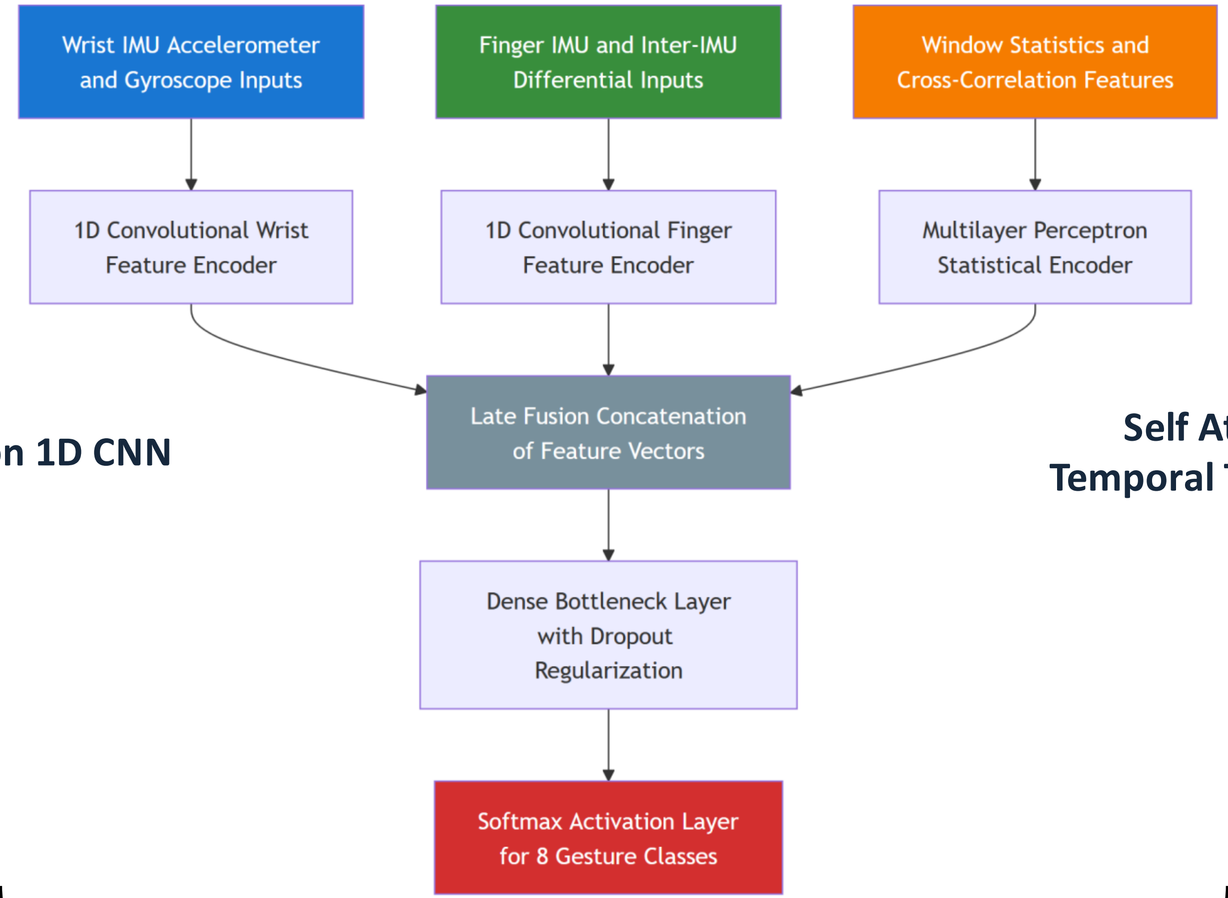


Gesture Recognition With IMU Data

Implementation: Model Architectures.

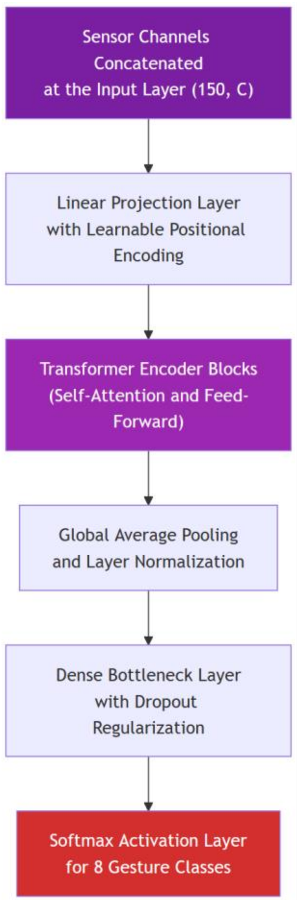


Early Fusion 1D CNN



Late Fusion (Multi Branch) 1D CNN

Self Attention Temporal Transformer



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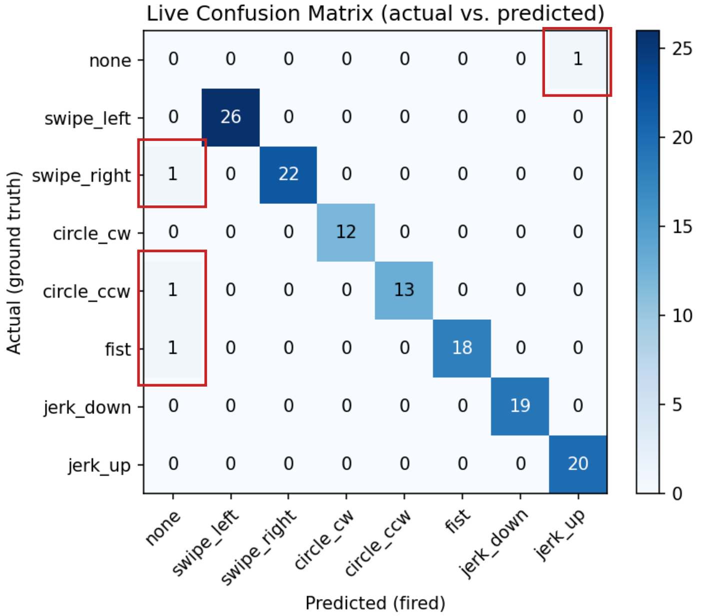
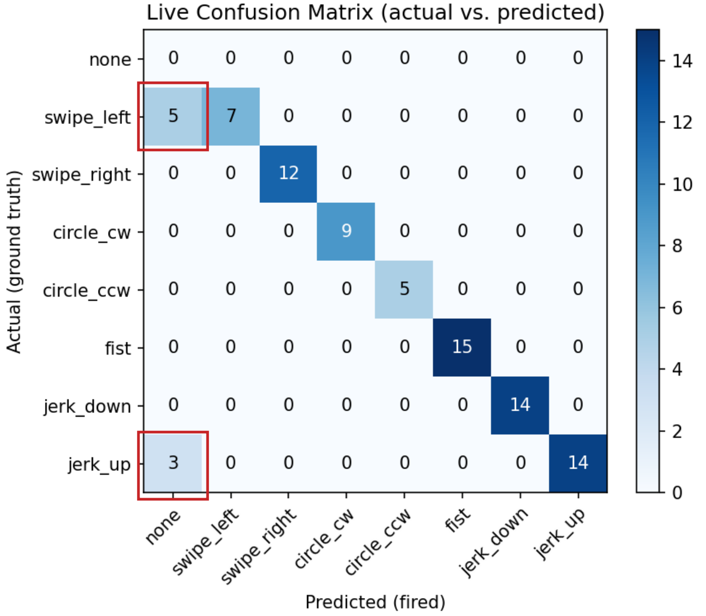
Gesture Recognition With IMU Data

Implementation: Model Architectures – Real World Performance.



Early Fusion 1D CNN



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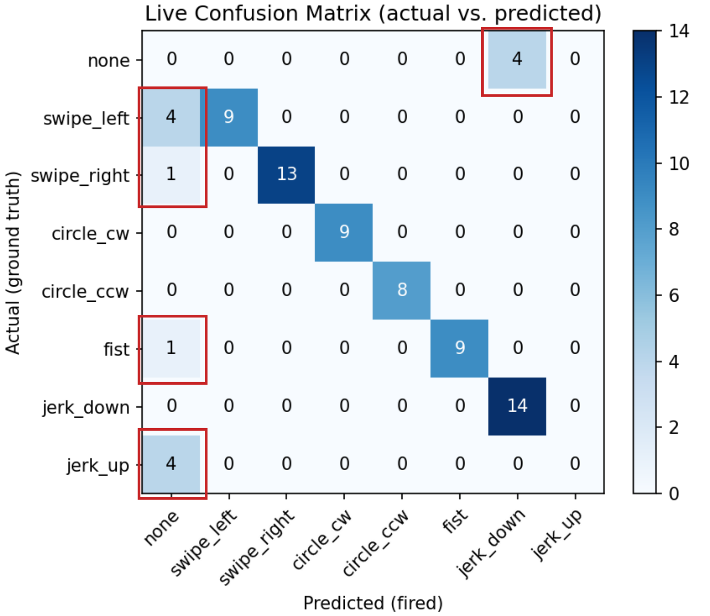
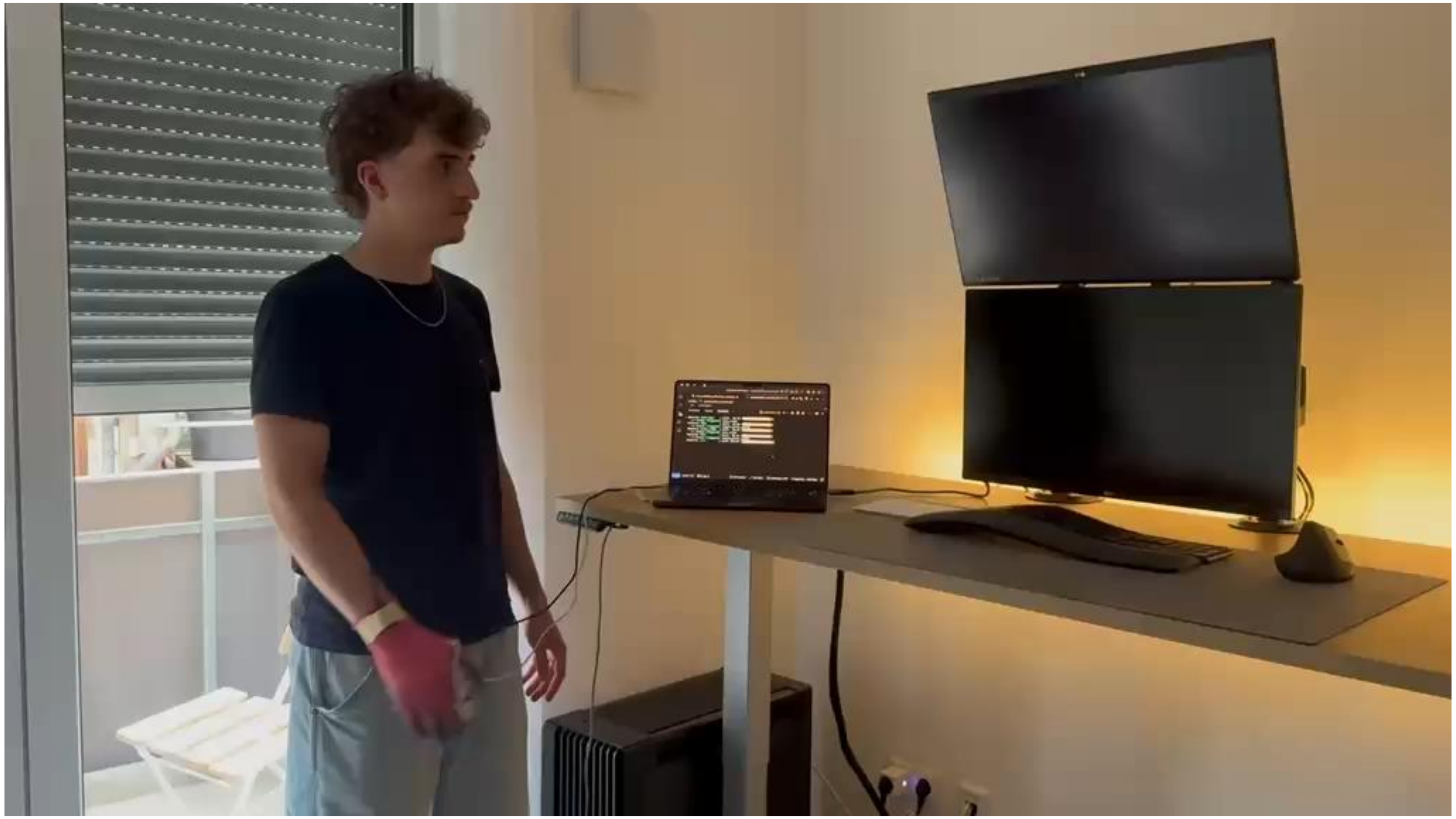


Self Attention Temporal Transformer

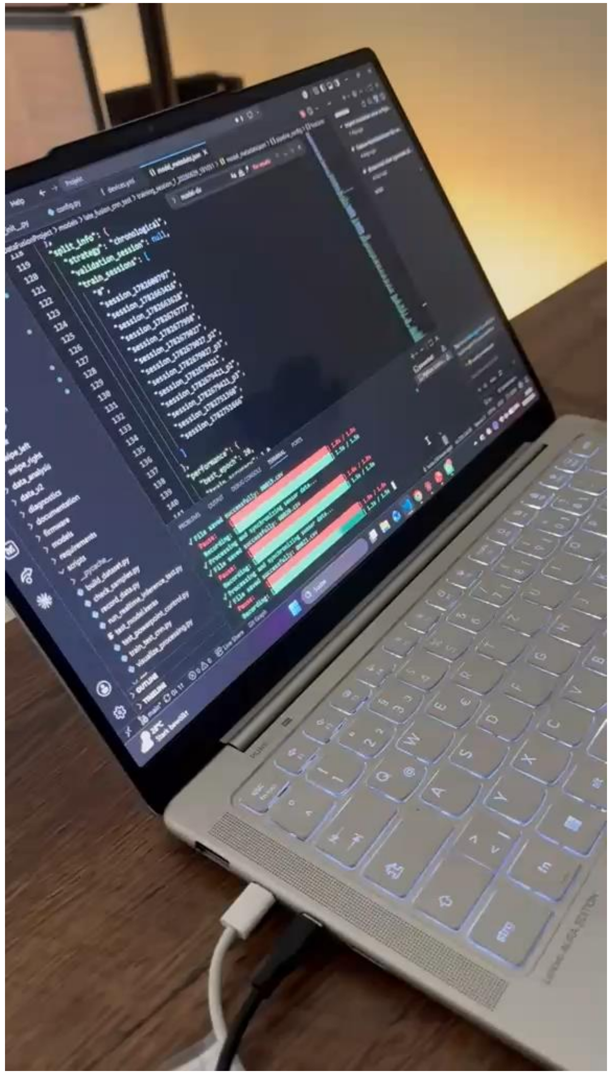
Gesture Recognition With IMU Data

Demonstration: Power Point Control.



Gesture Recognition With IMU Data

Demonstration: Data Recording.



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